0.1 Coupled Degenerate Darcy-Stokes System

We consider a degenerate Darcy-Stokes coupled boundary-value problem in the dimensionless form:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{2}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{3}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{4}$$

and

$$\begin{aligned}
\phi^{1+\Theta}\check{\mathbf{v}}_r\cdot\boldsymbol{\nu} &= g_r && \text{on } \Gamma_{u,D},\\
\check{\mathbf{v}}_s &= \mathbf{g}_s && \text{on } \Gamma_{u,S},\\
\check{q}_l &= \phi^{1/2}q_{l,0} && \text{on } \Gamma_{i,D},\\
\boldsymbol{\tau}\cdot\boldsymbol{\nu} - q &= \boldsymbol{t}_0 && \text{on } \Gamma_{i,S}.
\end{aligned} \tag{5}$$

Here, $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the parts of the boundary, where essential boundary conditions are prescribed for the Darcy and the Stokes problems, respectively, while $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the portions of the boundary, where natural boundary conditions are imposed. We assume the essential and natural boundary conditions are non overlapping and covering the entire boundary, i.e., $\Gamma_{u,D}\cup\Gamma_{i,D} = \partial\Omega$, $\Gamma_{u,D}\cap\Gamma_{i,D} = \emptyset$ and $\Gamma_{u,S}\cup\Gamma_{i,S} = \partial\Omega$, $\Gamma_{u,S}\cap\Gamma_{i,S} = \emptyset$.

To ensure well-posedness in the presence of degeneracy ($\phi = 0$), we must verity that all terms scaled by $\phi^{-1/2}$ remain well-defined. We expand the gradient term in equation (1) and the divergence term in equation (2) as

$$\begin{aligned}
\phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &= \phi^{1/2+\Theta}\nabla\check{q}_l + \phi^{\Theta-1/2}\check{q}_l\nabla\phi,\\
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) &= \phi^{1/2+\Theta}\nabla\cdot\check{\mathbf{v}}_r + \phi^{\Theta-1/2}\nabla\phi\cdot\check{\mathbf{v}}_r,
\end{aligned} \tag{6}$$

and they are well-defined provided that

$$\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^2. \tag{7}$$

We define a scaled Hilbert space tailored to the present formulation, within which the scaled velocity $\check{\mathbf{v}}_r$ should lie in.

Definition 0.1. Consider a scaled Hilbert space:

$$H_\phi(\mathrm{div};\Omega) = \Big\{\mathbf{v}\in(L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\in L^2(\Omega)\Big\}, \tag{8}$$

which is a Hilbert space equipped with the inner product

$$(\mathbf{u},\mathbf{v})_{H_\phi(\mathrm{div};\Omega)} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})). \tag{9}$$

The normal trace on $\partial\Omega$ is well-defined as

$$\left\|\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu}\right\|_{H^{-1/2}(\partial\Omega)} < C_\Omega\Big\{ \|\mathbf{v}\|_{L^2(\Omega)} + \left\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\right\|_{L^2(\Omega)} \Big\} \tag{10}$$

subsequently, we have the space $H_\phi^{-1/2}(\mathrm{div};\partial\Omega)$ understood as the image of the normal trace operator on $H_\phi(\mathrm{div};\Omega)$.

0.1.1 The weak formulation

We assume that the Dirichlet data $\mathbf{g}_s \in (H^{1/2}(\Gamma_{u,S}))^2$, and understand $\tilde{\mathbf{g}}_s \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. Similarly, we assume that $\phi^{-1/2}g_r \in H_\phi^{-1/2}(\partial\Omega)$ understood as the image of the scaled normal trace operator (10). Let

$$\phi^{1/2+\Theta}\mathbf{g}_r\cdot\boldsymbol{\nu} = \phi^{-1/2}g_r \quad \text{on} \quad \partial\Omega. \tag{11}$$

We view $\tilde{\mathbf{g}}_r$ as a continuous extension from the boundary into the domain.

The following energy spaces are defined for this scaled Darcy-Stokes coupled system

$$\begin{aligned}
\mathbb{V}_D^\phi &= \Big\{\mathbf{v}\in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\Big\},\\
\mathbb{U}_D^\phi &= \Big\{\mathbf{u}\in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{u}\cdot\boldsymbol{\nu} = \phi^{-1/2}g_r \text{ on } \Gamma_{u,D}\Big\} = \tilde{\mathbf{g}}_r + \mathbb{V}_D^\phi,\\
\mathbb{W}_D &= L^2(\Omega),\\
\mathbb{V}_S &= \Big\{\mathbf{v}\in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\Big\},\\
\mathbb{U}_S &= \Big\{\mathbf{u}\in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g}_s \text{ on } \Gamma_{u,S}\Big\} = \tilde{\mathbf{g}}_s + \mathbb{V}_S,\\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned} \tag{12}$$

where each space is equipped with their natural norms.

A compatibility condition is required to ensure overall mass conservation when essential boundary conditions are prescribed on the entire boundary $\partial\Omega$, i.e., when $\Gamma_{u,D} = \Gamma_{u,S} = \partial\Omega$. In this case, the prescribed Dirichlet data must satisfy the following compatibility condition:

$$\int_{\partial\Omega} (\phi^{1+\Theta}\mathbf{g}_r + \mathbf{g}_s)\cdot\boldsymbol{\nu}\, ds = 0. \tag{13}$$

In practice, we may not enforce this compatibility condition explicitly due to the pressence of natural boundary conditons.

The standard mixed form for the proposed coupled system reads as follows:

Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^{\phi}$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{14}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_S, \tag{15}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_r)) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{16}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_l) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_l\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_D. \tag{17}$$

We state the well-posedness result of the scaled formulation directly as follows.

Lemma 0.1. *(Existence and Uniqueness)*

Assume the condition (7) *holds,* $0 \le \phi \le \phi^* < 1$*. There exists a unique solution to the scaled formulation* (14)–(17)*, and it satisfies*

$$\begin{aligned} &\|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)\big\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2} \\ &\qquad \le C\big\{|\rho_r| + \|\tilde{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{g}}_r)\big\|_{(L^2(\Omega))^2} + \|\tilde{\mathbf{g}}_s\|_{H^1(\Omega)}\big\}. \end{aligned} \tag{18}$$

Proof. See Arbogast et al. (2017). □

0.1.2 A scaled mixed finite element method

We discretize the scaled mixed formulation using two pairs of finite element spaces: $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h} \subset H(\mathrm{div};\Omega) \times L^2(\Omega)$ for the Darcy part of the system and $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h} \subset (H^1(\Omega))^2 \times L^2(\Omega)$ for the Stokes part of the system. We recall the aforementioned inf-sup stable mixed finite element space pairs on each element $E \in \mathcal{T}_h$:

- Reduced $H(\mathrm{div})$ conforming space $\mathbb{V}_1^0$ in Section **??**,

$$\begin{aligned} \mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \mathrm{curl}\,\mathbb{S}_2^{\mathcal{DS}}(E), \\ \mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E); \end{aligned} \tag{19}$$

- Modified Bernardi-Raugel element in Section **??**,

$$\begin{aligned} \mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \\ \mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E). \end{aligned} \tag{20}$$

We impose essential boundary conditions by projecting extensions $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ into the finite element spaces as $\hat{\mathbf{g}}_r \in \mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s \in \mathbb{V}_{S,h}$.

We also define

$$\begin{aligned} \mathbb{V}_{D,h,0} &= \{\mathbf{v} \in \mathbb{V}_{D,h} : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\} \\ \mathbb{V}_{S,h,\mathbf{0}} &= \{\mathbf{v} \in \mathbb{V}_{S,h} : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\} \end{aligned} \tag{21}$$

The scaled mixed finite element method: Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h,0} + \hat{\mathbf{g}}_r$, $\check{q}_{r,h} \in \mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h} \in \mathbf{V}_{S,h,\mathbf{0}} + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_{S,h}$ such that

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla\boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}), \qquad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{22}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_h\Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{23}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{24}$$

$$(\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_{l,h}\Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{25}$$

The existence and convergence results of the proposed new mixed finite element discretization can be established in complete analogy with the proof in Arbogast et al. (2017). We state the results below without repeating the full derivation for completeness.

Lemma 0.2. *If* (7) *holds, then there exists a unique solution to the scaled mixed finite element method* (22)–(25).

Proof. See Arbogast et al. (2017). □

Lemma 0.3. *Assume that* (7) *holds on the porosity,* $0 \le \phi \le \phi^* < 1$, *the mesh is quasiuniform. Then the difference of the solution to the scaled formulation* (22)–(25), *and its finite elment approximation satisfy*

$$\begin{aligned}
&\|\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D,h}}\left[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}))\right]\right\|_{L^2(\Omega)} + \|\mathbf{v}_s - \mathbf{v}_{s,h}\|_{H^1(\Omega)} \\
&+ \|\tilde{q}_l - \tilde{q}_{l,h}\|_{L^2(\Omega)} + \|q - q_h\|_{L^2(\Omega)} \\
&\leq C\bigg\{ \left\|\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r\right\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D.h}}[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r))]\right\|_{L^2(\Omega)} \\
&+ \left\|\pi_{\mathbb{V}_1^0}\tilde{\mathbf{g}}_r - \hat{\mathbf{g}}_r\right\|_{L^2(\Omega)} + \left\|\tilde{q}_l - \mathcal{P}_{\mathbb{W}_{D,h}}\tilde{q}_l\right\|_{L^2(\Omega)} + \left\|q - \mathcal{P}_{\mathbb{W}_{S,h}}q\right\|_{L^2(\Omega)} \\
&+ \|\mathbf{v}_s - \pi_{\mathbb{V}_{\mathcal{BR}}}\mathbf{v}_s\|_{H^1(\Omega)} + \|\pi_{\mathbb{V}_{\mathcal{BR}}}\tilde{\mathbf{g}}_s - \hat{\mathbf{g}}_s\|_{H^1(\Omega)} \bigg\},
\end{aligned} \tag{26}$$

where $\pi_{\mathbb{V}_1^0}$ *and* $\pi_{\mathbb{V}_{\mathcal{BR}}}$ *are porjection operators defined in section* **??** *and section* **??** *respectively.*

Proof. See Arbogast et al. (2017). □

0.1.3 Local Mass Conservation

Before proceeding, we ensure that the formulation is well-defined—specifically, that no division by zero occurs. The term $\phi^{-1/2}$ in the symmetric equations (24) and (25) causes trouble when there is no melting, i.e., $\phi = 0$. To address this, we utilize an element-averaged variable ϕ_E as

$$\phi_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})\,d\mathbf{x}, \tag{27}$$

where $|E|$ is the area of element E and $\phi_E = \mathcal{P}_{\mathbb{W}_{D,h}}\phi|_E \in \mathbb{W}_{D,h}$. Then the two terms involving divergence is modified as

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})|_E = \begin{cases}\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) & \text{if } \phi_E \neq 0, \\ 0 & \text{if } \phi_E = 0,\end{cases} \tag{28}$$

where the point-wise term $\phi^{-1/2}$ is modified to be a cell-averaged term $\phi_E^{-1/2}$. We apply the divergence theorem to expand the integral term involving the divergence operator in equation (25) as

$$\int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_{l,h}\,d\mathbf{x} = \int_{\partial E}\phi_E^{-1/2}(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu})\omega_{l,h}\,ds, \tag{29}$$

where $\omega_{l,h} \in \mathbb{W}_{D,h}$ is constant on the element $E \in \mathcal{T}_h$. We now propose our local mass conserved version with $\phi_E^{1/2}$ as

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \qquad \forall \boldsymbol{\psi}_{s,h}, \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{30}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi \phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h), \omega_h\Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{31}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{32}$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{\phi \phi_E^{-1}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h), \omega_{l,h}\Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{33}$$

Here, the term $\phi\phi_E^{-1}$ in equation (33) is viewed as 1 when $\phi_E = 0$. To verify local mass conservation of the fluid, consider any element $E \in \mathcal{T}_h$, equation (33) gives

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h})\, d\mathbf{x} + \int_E \frac{\phi}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h)\, dx = 0, \tag{34}$$

and equation (31) gives

$$\int_E \nabla \cdot \mathbf{v}_{s,h}\, d\mathbf{x} - \int_E \frac{\phi}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h)\, dx = 0. \tag{35}$$

We sum equations (31) and (33) together assuming that $\omega_h = \omega_E \in \mathbb{W}_{S,h}$ and $\omega_{l,h} = \phi_E^{-1/2}\omega_E \in \mathbb{W}_{D,h}$. The local mass conservation of the phase-averaged mixture is now retrieved as

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r + \check{\mathbf{v}}_s)\, d\mathbf{x} = 0. \tag{36}$$

Lemma 0.4. *Assuming* (7) *holds, there exists a unique solution to the modified local mass conservation modified scaled mixed finite element method* (30)–(33).

Proof. See Arbogast et al. (2017). The local mass conservation modification replace the term $1/(1-\phi)$ with $(\phi\phi_E^{-1})/(1-\phi)$,which makes it completely analogous to the previous proof. Hence, we are not repeating here. □

To derive a bound for the error, we take the difference of the scaled formulation (14)–(17) and the discretized formulation with local mass conservation modification (30)–(33):

0.2 Saddle Point System

Saddle point system arises frequently in scientific and engineering applications, including constrained optimization problem, image reconstruction and optimal control. Saddle point system has been extensively and systematically studied, notably by Benzi et al. (2005). In our case, the saddle point system arises naturally from the mixed finite element discretization.

In this section, we give a brief overview of the properties of a saddle point systems and present an iteration method tailored to our problem. We express the abstract saddle point system as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{37}$$

where we assume the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

0.2.1 Reduced system

We now present a detailed discussion of the saddle point system (SPS) arising from the mixed finite element method (MFEM) including the treatment of boundary conditions.

As a first step, we regroup the degrees of freedom (dofs) ($i \in I$) into three distinct sets: the set of the interior dofs I_c, the set of the boundary dofs assigned with essential boundary condition I_{Γ_u} and the set of the boundary dofs assigned with natural boundary condition I_{Γ_i}. We have $I_c \cap I_{\Gamma_u} = I_c \cap I_{\Gamma_i} = I_{\Gamma_u} \cap I_{\Gamma_i} = \emptyset$ and $I_c \cup I_{\Gamma_i} \cup I_{\Gamma_u} = I$. We then reorganize the $\mathbf{A}$ matrix accordingly as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{38}$$

where $\mathbf{A}_K = \{a_{k_1,k_2}\}$ with $k_1, k_2 \in I_c + I_{\Gamma_i}$, consists of all the "free" dofs. The matrix $\mathbf{A}_M = \{a_{m_1,m_2}\}$ with $m_1 \in I_c + I_{\Gamma_i}$, $m_2 \in I_{\Gamma_u}$, represents the interaction between "free" and prescribed dofs. Finally, $\mathbf{M} = \{m_{i,j}\}$, $i, j \in I_{\Gamma_u}$ accounts for the contributions from the fixed dofs associated with essential boundary conditions.

We apply the same treatment to the matrix $\mathbf{B}$ and the right-hand side vector $\mathbf{F}$ as

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix} \text{ and } \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix}. \tag{39}$$

We reformulate the linear system using the regrouped dofs and derive a reduced system of the form:

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M \mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T \mathbf{g} \end{bmatrix}, \tag{40}$$

where $\mathbf{g}$ denotes the values prescribed by the essential boundary condition, and $\mathbf{g}_N$ represents the contribution from the natural boundary conditions. The reformed reduced system remains the structure of a saddle point system, and can be represented using the same abstract form (37).

0.2.2 Linear system for coupled Darcy-Stokes equations

The linear system formed from the equations (30)–(33) is shown as

$$\begin{bmatrix} \mathbf{A}_S & -\mathbf{B}_S & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_S^T & \mathbf{C}_S & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_D & -\mathbf{B}_D \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_D^T & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{y}_S \\ \mathbf{x}_D \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{G}_S \\ \mathbf{F}_D \\ \mathbf{G}_D \end{bmatrix}, \tag{41}$$

wherein all the matrices are reduced matrices excluding boundary dofs associated with essential boundary conditions, and all the right-hand side vectors are modified by boundary conditions. Most of these matrics can be computed directly from the definition of the inner product over each element E. However, the $\mathbf{B}_D$ matrix should be computed using the divergence theorem (29) to avoid evaluating the derivatives of the porosity ϕ. For any element $E \in \mathfrak{T}_h$,

$$\begin{aligned} \mathbf{B}_D &= \left(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}\right) \\ &= \int_e (\phi_E^{-1/2} \phi^{1+\Theta}) \check{\mathbf{v}}_{r,h} \cdot \boldsymbol{\nu} \omega_{l,h} \, ds, \end{aligned} \tag{42}$$

where the term $\phi_E^{-1/2} \phi^{1+\Theta}$ is well-defined and evaluates 0 if $\phi_E = 0$.

We then form a double saddle point system by symmetrically permutating the original coupled linear system as

$$\begin{bmatrix} \mathbf{A}_S & \mathbf{0} & -\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D & \mathbf{0} & -\mathbf{B}_D \\ \mathbf{B}_S^T & \mathbf{0} & \mathbf{C}_S & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_D^T & \mathbf{K} & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{x}_D \\ \mathbf{y}_S \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{F}_D \\ \mathbf{G}_S \\ \mathbf{G}_D \end{bmatrix}. \tag{43}$$

We group these block matrices accordingly and reterive the simple abstract saddle point system given in equation (37),

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_D \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{C}_S & \mathbf{K} \\ \mathbf{K} & \mathbf{C}_D \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{F}_D \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{G}_S \\ -\mathbf{G}_D \end{bmatrix}, \tag{44}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}. \tag{45}$$

We now examine the structure of the resulting saddle point system. The scaled formulation (16) guarantees that the matrix $\mathbf{A}_D$ is nonsingular, therefore the coupled $\mathbf{A}$ is nonsingular as well. The saddle point system can be factorized as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} = \begin{bmatrix} I & \mathbf{O} \\ \mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S} \end{bmatrix} \begin{bmatrix} I & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & I \end{bmatrix}, \tag{46}$$

where $\mathbf{S}$ is the Schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T),$$

The presence of an evolving no-melting regions, where the porosity $\phi = 0$, poses some obstacle to some traditional solver. On one hand, the existence of zero porosity region guarantees that $\mathbf{B}$ matrix formed in Darcy system does not has full column rank. is usually not invertable directly. On the other hand, in addition to the constant null space induced by gradient of pressure, the null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of zero porosity region).

Taking into consideration that the solver will be implemented in parallel, a iterative solver is more suitable than direct solver.

0.2.3 Schur Complement

To begin with, we discuss a "direct" way to solve this derived saddle point system. Now consider a general saddle point system (37) with symmetric positive-definite matrix $\mathbf{A}$ as

$$\begin{aligned} \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} &= \mathbf{f}, \\ \mathbf{B}^T\mathbf{x} - \mathbf{C}\mathbf{y} &= \mathbf{g}. \end{aligned} \tag{47}$$

We represent the solution vector $\mathbf{x}$ as

$$\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{B}\mathbf{y}), \tag{48}$$

where $\mathbf{A}^{-1}$ is computed with conjugate gradient algorithm. We then eliminate $\mathbf{x}$ to represent $\mathbf{y}$ as

$$\mathbf{y} = \mathbf{S}^{-1}(\mathbf{g} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{f}), \tag{49}$$

where $\mathbf{S}^{-1}$ is computed with Minimal Resigual algorithm Paige and Saunders (1975) due to the formed Schur complement being a symmtric indefinite system.

0.2.4 Uzawa Iteration

Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), which gained popularity in computational fluid dynamics. It has later been applied to the case of discretizing Stokes problem with Mixed Finite Element. Some inexact inner solver to accelerate uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We start by stating an abstract inexact ezawa iteration algorithm in Algorithm 1. Some choose diagonal matrix $\alpha\mathbf{I}$ for precondictioner $\mathbf{Q}_A$. However, such

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$

6: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

7: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$

8: **end while**

simple matrix wouldn't generate a convergence result when applied to our coupled problem. In the next section, we introduce a modified algorithm that would solve our system.

0.2.5 Modified Uzawa Iteration

We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system.

Algorithm 2 Modified Uzawa Iteration

Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.
while res $>$ tol **do**
  Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
  Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$
  Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
  Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
  Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
  Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$
end while

0.3 Convergence Test

We test our MFEM setup with two different scenarios. First, we test a pseudo 1D problem but with 2D elements equipped with three different porosity scenarios. Second, we test a more complex yet a more realistic problem representing Mid Ocean Ridge where no true solution is presented.

In this subsection, we perform a convergence test on a 2D grid with a pseudo 1D problem. We study a compacting column in one direction. The column extends over $y \in [-L, L]$ and has no flow through top and bottom boundaries. We prescribe the following boundary conditions

- Bottom Edge;
  - Essential boundary condition:

$$\begin{cases} \check{\mathbf{v}}_s = \mathbf{0}; \\ \mathbf{b} = \mathbf{0}; \end{cases}$$

  - Essential boundary condition: Normal Darcy flux $\check{\mathbf{v}}_r \cdot \boldsymbol{\nu} = 0$;
- Left and Right side Edges;
  - y component of Stokes velocity is free, x component is kept as zero;
  - Darcy velocity, which is understood as flux is also kept zero;
  - normal flux dof fixed with a zero flux.

- Top Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.

We assume no velocity in x direction, $\check{\mathbf{v}}_s \cdot \mathbf{i} = 0$, and $\check{\mathbf{v}}_r \cdot \mathbf{i} = 0$, where $\mathbf{i}$ is the unit vector in x direction pointing from left to right. Let $u = \phi \check{\mathbf{v}}_r \cdot \mathbf{j}$ and $v = \check{\mathbf{v}}_s \cdot \mathbf{j}$, where $\mathbf{j}$ is the unit vector in y direction pointing upwards.

$$\begin{aligned} u + \phi^2 q_l' &= 0, \\ u' + \phi(q_l - q_s) &= 0, \\ [q_s - \frac{1}{3}(1-4\phi)v']' &= 1 - \phi, \\ v' - \phi(q_f - q_s) &= 0, \end{aligned} \tag{50}$$

When no melting presents $\phi = 0$, the equations reduce to

$$u = -v = 0, \quad \text{and} \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{51}$$

When melting presents $\phi > 0$, the equations can be reduced to

$$\phi^{2+2\Theta}\left(\frac{3+\phi-4\phi^2}{3\phi}u'\right)' - u = \phi^{2+2\Theta}(1-\phi), \tag{52}$$

where all other quantities can be determined according to u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2-2\Theta}u, \quad \text{and} \quad q_s = u'/\phi + q_l. \tag{53}$$

When ϕ is constant, we define an auxiliary function R as

$$R = \left(\phi^{2+2\Theta}\frac{3+\phi-4\phi^2}{3\phi}\right)^{-1/2}. \tag{54}$$

We test our MFEM code on three porosity settings, and use the closed form solutions derived in Arbogast et al. (2017).

0.3.1 Column compaction with Constant porosity

$$\phi_0(z) = \phi_0; \tag{55}$$

Closed form solution:

$$
\begin{aligned}
u &= -\phi_0^2(1-\phi_0)\left[1-\frac{\cosh(Rz)}{\cosh(RL)}\right],\\
v &= -u,\\
q_l &= (1-\phi_0)\left[z-\frac{1}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right]+c,\\
q_s &= (1-\phi_0)\left[z-\frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right]+c,
\end{aligned}
\tag{56}
$$

where c is a constant kernel.

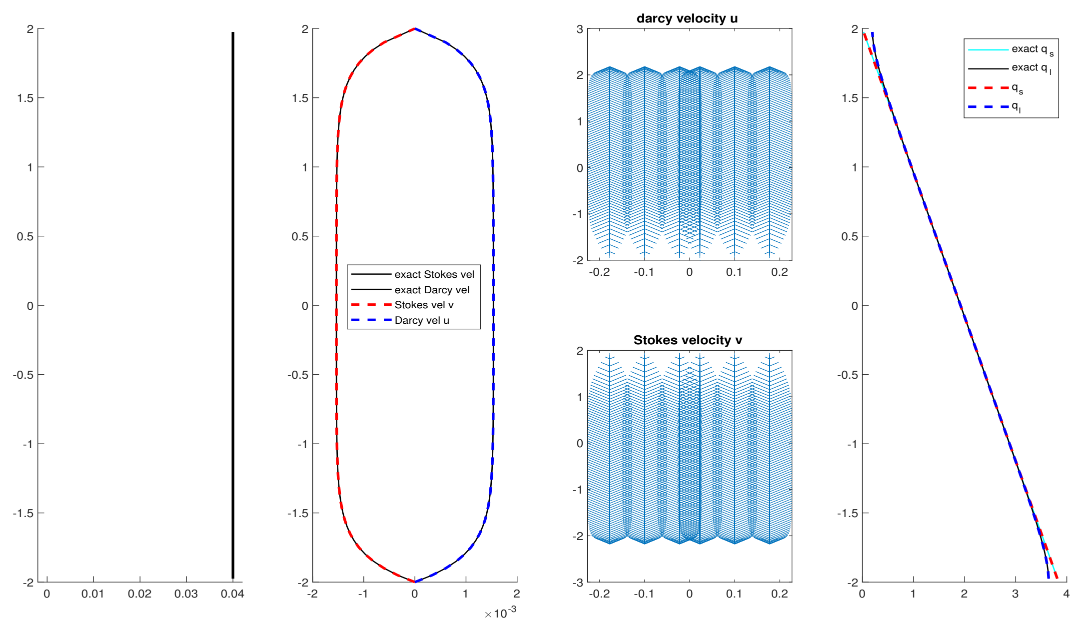

Figure 1: Constant porosity compaction column test.

0.3.2 Piecewise constant porosity

$$
\phi_J(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 & \text{if } z \leq 0, \end{cases} \tag{57}
$$

Closed form solution:

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	3.285e-03	-	3.638e-05	-
2×40	8.047e-06	1.95	8.047e-06	1.95	8.731e-04	1.91	9.671e-06	1.91
2×80	2.031e-06	1.99	2.031e-06	1.99	2.218e-04	1.98	2.457e-06	1.98
2×160	5.091e-07	2.00	5.091e-07	2.00	5.567e-05	1.99	6.166e-07	1.99

Table 1: Constant porosity compaction column test.

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	3.794e-03	-	4.202e-05	-
2×40	7.681e-06	1.96	7.681e-06	1.96	1.005e-03	1.92	1.114e-05	1.92
2×80	1.958e-06	1.97	1.958e-06	1.97	2.472e-04	2.02	2.738e-06	2.02
2×160	4.974e-07	1.98	4.974e-07	1.98	5.972e-05	2.04	6.615e-07	2.05

Table 2: Piecewise Constant porosity compaction column test.

We conclude that $u = -v = 0$ when $z > 0$, and $q_l = 0$, $q_s = z + c$. When $z < 0$, $\phi_0 \neq 0$,

$$
\begin{aligned}
u &= \phi_0^2(1-\phi_0)\left[1 - \cosh(Rz) + \frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Rz)\right],\\
v &= -u,\\
q_l &= -(1-\phi_0)\left[z - \frac{1}{R}\sinh(Rz) + \frac{1-\cosh(RL)}{R\sinh(RL)}\cosh(Rz)\right] + c,\\
q_s &= q_l + \frac{u'}{\phi_0} + c,
\end{aligned}
\tag{58}
$$

where c is a constant kernel shifting the pressure potential.

0.3.3 Column compaction with quadratic porosity.

$$
\phi_2(z) = \begin{cases} 0 & \text{if } z > 0,\\ \phi_0 z^2 & \text{if } z \leq 0. \end{cases} \tag{59}
$$

We are interested in presenting the result when $\phi_0 = 0.01$, which is not approximated precisely using previous closed form solution due to the assumption $\phi \ll 1$ does not hold anymore. Here, we present an alternative way of constructing approximation solution of the second order ordinary differential equation (52). Let $v = u/\phi^2 =$

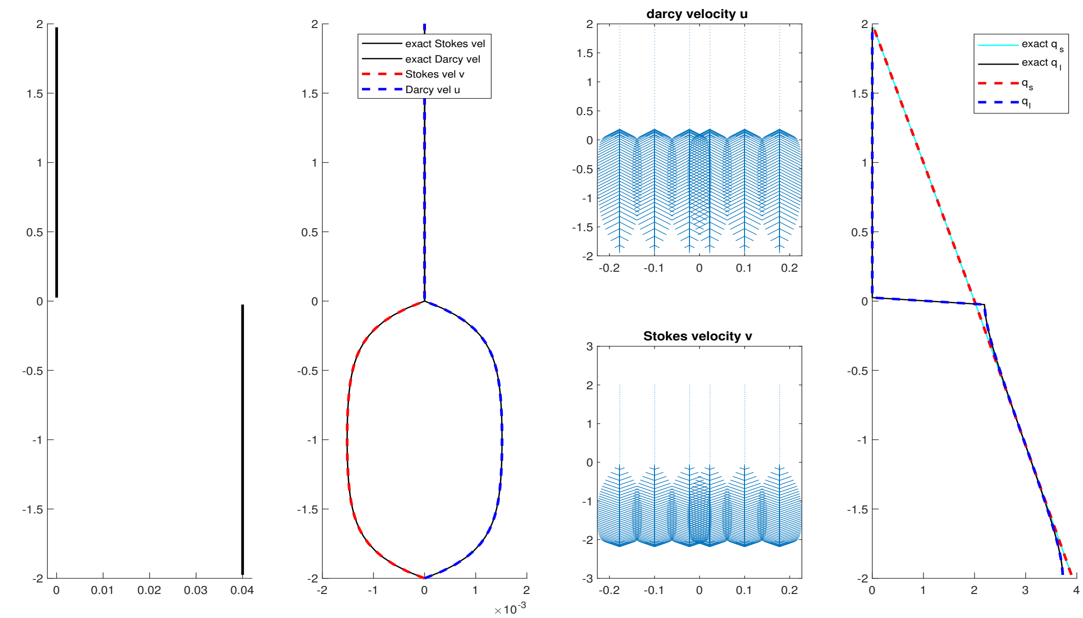

Figure 2: Piecewise constant porosity compaction column test.

$u/(\phi_0^2 z^4)$,

$$v' = \frac{-4v}{z} + \frac{u'}{\phi_0^2 z^4}. \tag{60}$$

Substitute u with v, we have

$$f(u'', u', u, z) = \left[\frac{3 + \phi_0 z^2 - 4\phi_+ z^4}{3\phi_0 z^2}\left(\phi_0^2 z^4 v' + 4\phi_0^2 z^3 v\right)\right]' - v = 1 - \phi_0 z^2. \tag{61}$$

We assume that there exists a solution that can be represented as a sum of an infinity series. The solution can be understood as the sum of a particular solution and a homogeneous power-series solution, which solves the homogeneous second order ODE $f(u'', u', u', z) = 0$. We apply Frobenius method here, assuming the homogeneous solution has the form as

$$v_H = (z - z_0)^r \sum_{n=0}^{\infty} a_n (z - z_0)^n, \tag{62}$$

where we pick $z_0 = 0$ for simplicity. We write the homogeneous recurrence as

$$
\begin{aligned}
&\left\{\phi_0\Big[(n+r-1)(n+r)+6(n+r)+4\Big]-1\right\}a_n+\\
&\left\{\frac{\phi_0^2}{3}\Big[(n+r-2)(n+r+3)+4+2(n+r+2)\Big]\right\}a_{n-2}+\\
&\left\{\frac{4\phi_0^3}{3}\Big[(n+r-4)(n+r+1)+4+4(n+r)\Big]\right\}a_{n-4}=0,
\end{aligned}
\tag{63}
$$

where we obtain the indicial equation be equating the coefficient associated to a_0 term to zero, i.e.,

$$\phi_0\big[(0+r-1)(0+r)+6(0+r)+4\big]-1=0, \tag{64}$$

where we solve for the positive root as

$$r=\frac{-5+\sqrt{9+4/\phi_0}}{2}. \tag{65}$$

Notice that odd and even coefficients are independent against each other, we can decompose the homogeneous solution into two independent power-series according to even and odd coefficients as

$$v_1 = z^r\sum_{n=0}^{\infty}a_{2n}z^{2n}, \quad v_2 = z^r\sum_{n=0}^{\infty}a_{2n+1}z^{2n+1}. \tag{66}$$

Then we proceed to determine the particular solution that solves the non-homogeneous function $f(u'', u', u, z) = 1 - \phi_0 z^2$.

$$v_p = \sum_{n=0}^{\infty}c_n z^n. \tag{67}$$

Substituting the particular solution into the non-homogeneous ODE (61), we obtain the coefficient by matching the coefficients on the right hand side according to different orders.

$$
\begin{aligned}
c_0 &= \frac{1}{4\phi_0-1}\\
c_2 &= \frac{-\phi_0-4\phi_0^2c_0}{18\phi_0-1}\\
c_4 &= \frac{80/3\phi_0^3c_0-10\phi_0^2c_2}{40\phi_0-1}.
\end{aligned}
\tag{68}
$$

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	9.558e-07	-	9.558e-07	-	3.088e-03	-	4.660e-06	-
2×40	3.176e-07	1.59	3.176e-07	1.59	1.525e-03	1.02	2.196e-06	1.09
2×80	8.686e-08	1.87	8.686e-08	1.87	7.442e-04	1.04	1.130e-06	0.96
2×160	2.222e-08	1.97	2.222e-08	1.97	2.893e-04	1.36	5.578e-07	1.02

Table 3: Quadratic porosity compaction column test $\phi_0 = 0.001$.

The rest of the coefficients can be determined with the recurrence form (63). Omitting the power-series v_2 containing only odd coefficient, we have the final form of the series solution as

$$v = v_p + C v_1, \tag{69}$$

and we determine the costant C with boundary condition $v(-2) = 0$, i.e., $C = -v_p(-2)/v_1(-2)$.

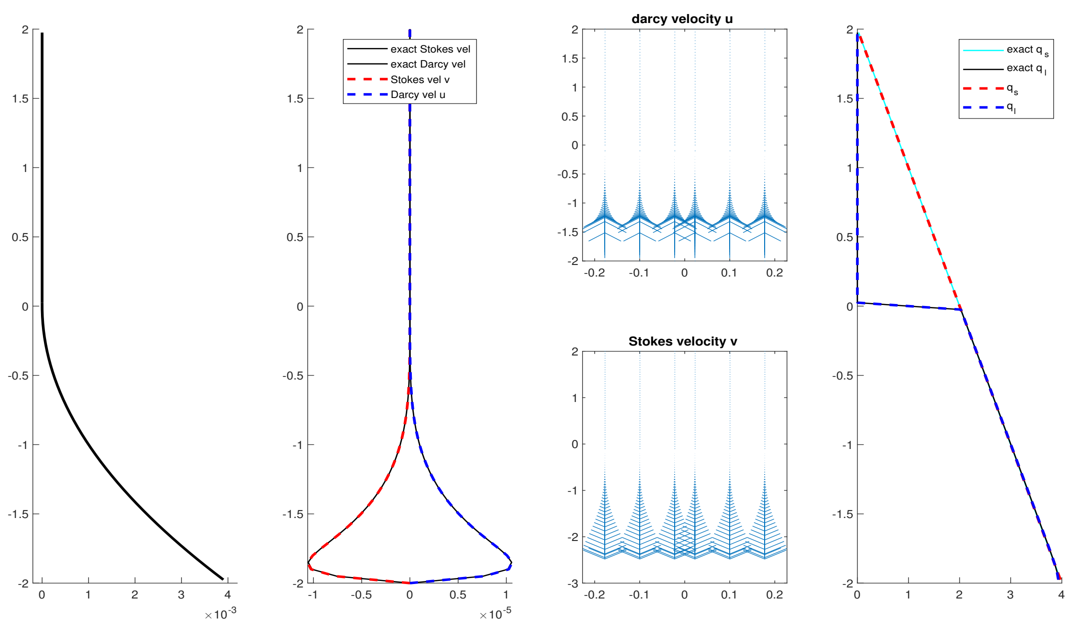

Figure 3: Quadratic porosity compaction column test $\phi_0 = 0.001$.

Now we redo the convergence test $\phi = \phi_0 z^2$ Since we do not make the assumetion that $\phi_0 z^{\ll} 1$ anymore, we test convergence on the case where $\phi_0 = 0.1$.

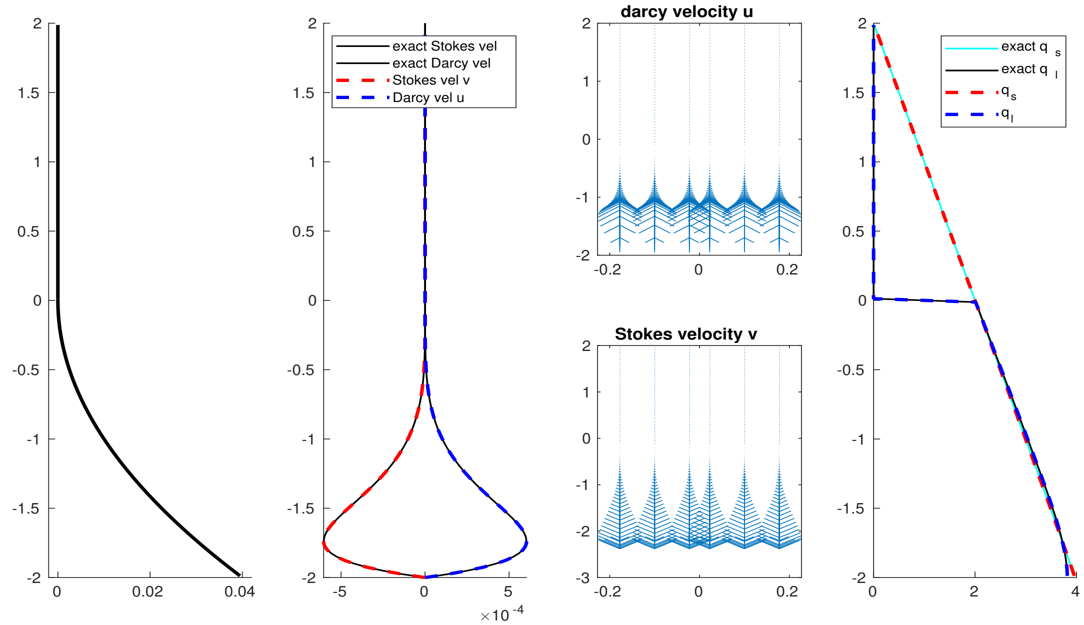

Figure 4: Quadratic porosity compaction column test $\phi_0 = 0.01$.

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.438e-05	-	2.438e-05	-		-		-
2×40	6.249e-06	1.96	6.249e-06	1.96				
2×80	1.486e-06	2.07	1.486e-06	2.07				
2×160	3.908e-07	1.96	3.908e-07	1.96				

Table 4: Quadratic porosity compaction column test $\phi_0 = 0.01$.

0.4 Mid-Ocean Ridge Test

We now reconsider the Mid-Ocean ridge test case in Arbogast et al. (2017). We solve the full system of equations on a dimensionless rectangular domain $x \in [-0.5, 0.5]$ and $z \in [-0.5, 0.0]$. The boundary conditions are imposed by assigning classical corner flow solution of Stokes velocity in Spiegelman and McKenzie (1987) as essential boundary condition all around the computational domain

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2+y^2)} \begin{pmatrix} \tan^{-1}(\frac{x}{-y})(x^2+y^2)+xy \\ z^2 \end{pmatrix}, \tag{70}$$

where $U_0 = 10^{-9}$m/s ≈ 3.2cm/yr is the constant upwelling velocity of mantle. According to mass conservation condition, we set no Darcy flux across the boundary.

We then set a not identically vanishing porosity by the formula

$$\phi(x,z)=\begin{cases}0.05(\dfrac{0.375+y}{0.375})^2(1-\dfrac{|x|}{|y|+l}), & \text{if } |y|\leq 0.375 \text{ and } |x|\leq z+l,\\ 0, & \text{otherwise,}\end{cases} \tag{71}$$

where $l = 20\,\mathrm{m}$ is the offset set to avoid singularity computation at the center $x = 0$ we translate the coordinates x to $x-l$ if $x<0$, and $x+l$ when $x>0$. In Figure 5, we show the porosity distribution in the computational domain. Unlike the previous test conducted in the quarter plane where $x \in (0, 0.5]$, $y \in [-0.5, 0.0]$. We perform the numerical test within the entire plane instead.

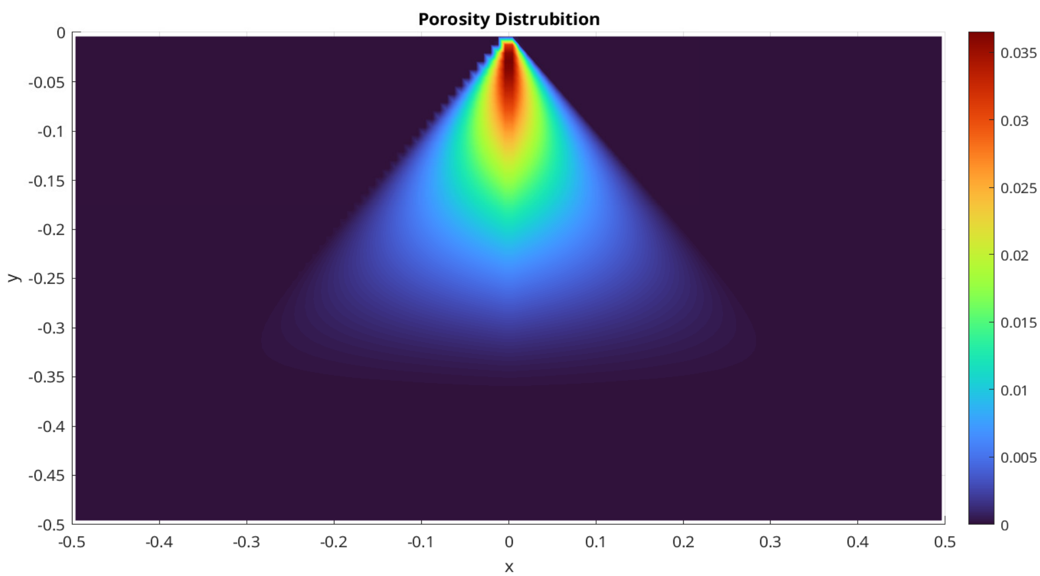

Figure 5: Porosity distribution.

In Figure 6 and Figure 7, we exhibit the nondimensional Stokes velocity $(\check{\mathbf{v}}_s, \check{q}_s)$ and Darcy velocity $(\phi\check{\mathbf{v}}_r, \check{q}_l)$ respectively. We observe the molten magma rise to the surface and focus to the region where most porosity presents. The solid matrix upwell from the bottom of the domain. We observe compaction at degree of melting (some porosity) where the buoyancy can no longer support the solid matrix on top of it due to density difference between solid and liquid. Despite of the presence of large no melting region, the system is solved successfully with modified Uzawa iteration

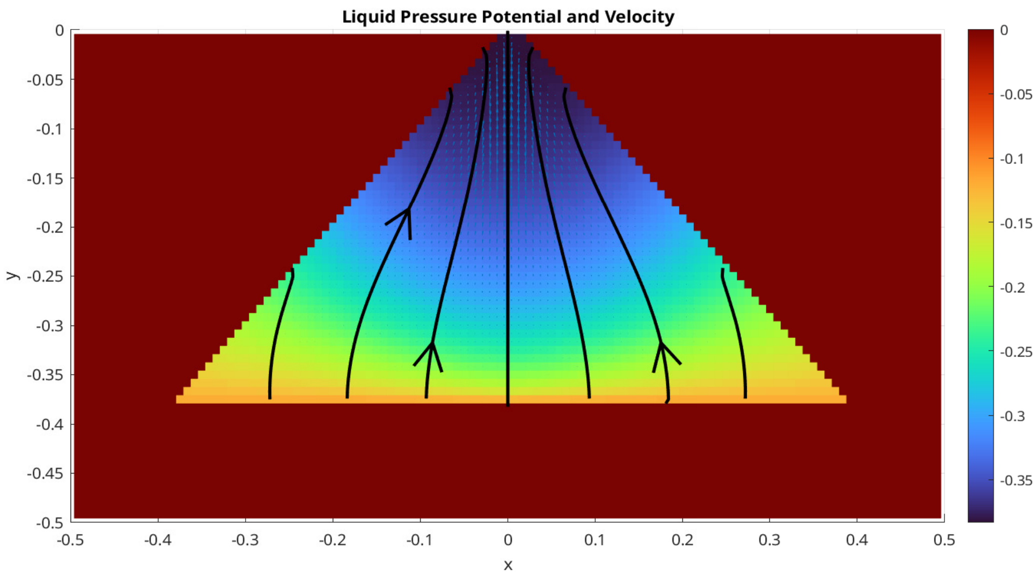

Figure 6: Liquid pressure potential and velocity.

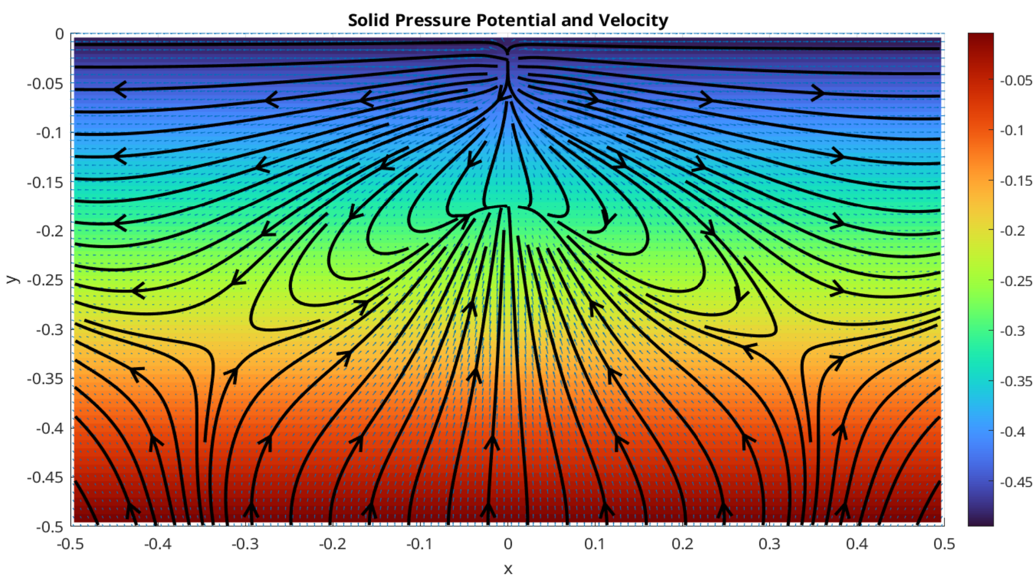

Figure 7: Solid pressure potential and velocity.

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